Ali Umar

(403) 615-8780 | aumar@andrew.cmu.edu | linkedin.com/in/aliumar10/

Education

Carnegie Mellon University — Pittsburgh, PA
Master of Science in AI Engineering – Biomedical Engineering — Dec. 2027

McMaster University — Hamilton, Canada
Bachelor of Engineering in Mechatronics and Biomedical Engineering — May 2026

Experience

McMaster University — Hamilton, Canada
Research Assistant — Sep. 2025 – May 2026

- Integrated **16** PCB sensor chips with **64** electrodes and Raspberry Pi via I2C interfacing, implementing signal amplification, filtering, and noise removal for real-time biomarker acquisition
- Developed regression-based pipeline to evaluate **24-hour** continuous aptamer sensing, standardizing feature extraction across partial and full scan intervals to optimize downstream assessment
- Overhauled legacy potentiostat code into modular architecture, accelerating setup time by **20%** and future-proofing codebase for multi-purpose project expansion

Spartan Controls — Calgary, Canada
Automation Specialist Co-op — May 2024 – Aug. 2025

- Built and configured **3** DeltaV control panels with **12** CHARMs each, integrating **120** I/O and **150** operator graphics for real-time process monitoring
- Engineered control logic sequences for a wastewater facility processing **100 million** liters/day for **250,000** residents, commissioning a new waste-gas burner across **4** control loops
- Authored custom Python verification functions within framework to perform module authentication, cutting downstream validation errors and boosting tool efficiency by **15%**

McMaster University — Hamilton, Canada
Software Research Assistant — May 2023 – May 2024

- Analyzed KMP and Boyer-Moore pattern matching algorithms in C leveraging power-of-two and prime encoding schemes, demonstrating a **25%** performance difference relevant to DNA/protein sequence analysis
- Architected a Python benchmarking framework to execute and evaluate **8** custom scripts across **1,000** automated test runs, streamlining runtime performance and eliminating manual testing overhead
- Initiated repository documentation and code refactoring to improve readability, while **synthesizing** experimental metrics into technical reports and presentations to brief team on progress and bottlenecks

Projects

3D Surgical Reconstruction from Endoscopic Video – *McMaster, Capstone Project* — Sep. 2025 – May 2026

- Implemented COLMAP and Neural Radiance Field reconstruction pipeline for endoscopic video, improving surgical margin accuracy to **90%**, minimizing resection risks and enhancing long-term patient outcomes
- Calibrated an IMU against checkerboard-based visual measurements, aligning inertial and camera data for multisensory navigation within an **8** mm endoscope constraint
- Characterized reconstruction accuracy using 3D-printed surgical cavity models and simulated test environments, with less than **3** mm reconstruction error and within **1** mm reproducibility

Rotator Cuff Rehab System – *McMaster, Course Project* — Jan. 2024 – May 2024

- Constructed a Bluetooth wearable utilizing ESP32 and BNO055 sensor in custom 3D enclosure, filtering IMU noise to capture objective joint kinematics during shoulder rehabilitation
- Introduced a real-time 3D interface comparing ideal vs. actual movements across joint angle, velocity, and acceleration, achieving **85%** accuracy and storing session data for clinician review

Skills

Programming Languages | Python, C/C++, MATLAB, SQL, Assembly, Verilog
Software Tools | Git, Linux, Simulink, ROS, Autodesk Inventor, VS Code, Quartus
Embedded Systems | Arduino, Microcontrollers, FPGA Systems, Oscilloscopes, Soldering